Ver23 PLC symbols and components

PCSCHEMATIC Automation

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Diagrams

-K1		001.001.001.001	-K2		001.001.001.002
I.00 /9.4 DI	ALARM_ON_1 Alarm button activated 1	X0	DI0 /9.7 DI Sink	+	X0 0
I.01 /9.5 DI	ALARM_ON_2 Alarm button activated 2	X1	DI1 /9.8 DI Sink	+	X1 1
I.02 DI	ALARM_ON_3 Alarm button activated 3	X2	DI2 DI Sink	+	X2 2
I.03 DI	ALARM_ON_4 Alarm button activated 4	X3	DI3 DI Sink	+	X3 3
I.04 DI	ALARM_ON_5 Alarm button activated 5	X4	DI4 DI Sink	+	X4 4
I.05 DI	ALARM_ON_6 Alarm button activated 6	X5	DI5 DI Sink	+	X5 5
I.06 DI	ALARM_ON_7 Alarm button activated 7	X6	DI6 DI Sink	+	X6 6
I.07 DI	ALARM_ON_8 Alarm button activated 8	X7	DI7 DI Sink	+	X7 7

24 VDC 0 VDC 24 VDC 0 VDC

+24V +24V /5.1

0V 0V /5.1

PE PE /5.1

You place the reference symbol by selecting Place all in the symbol section.
Add Plc - net - node - slot on the Symbol datafield tab to get a PlcID.
Give the component a name and place it.

All PLCs are pre-addressed from the database - Digital addresses are DI/DO and analog addresses are AI/AO.
Some components are defined with a statusype, which shows on the ref.symbol and on the IO-symbol.

The IO-list is imported to K1.

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-K3	001.001.001.003	-K4	001.001.001.004
DO0 DO	X0	O0 DO Relay	Y0 0
DO1 DO	X1	O1 DO Relay	Y1 1
DO2 DO	X2	O2 DO Relay	Y2 2
DO3 DO	X3	O3 DO Relay	Y3 3
DO4 DO	X4	O4 DO Relay	Y4 4
DO5 DO	X5	O5 DO Relay	Y5 5
DO6 DO	X6	O6 DO Relay	Y6 6
DO7 DO	X7	O7 DO Relay	Y7 7

24 VDC 0 VDC 24 VDC 0 VDC

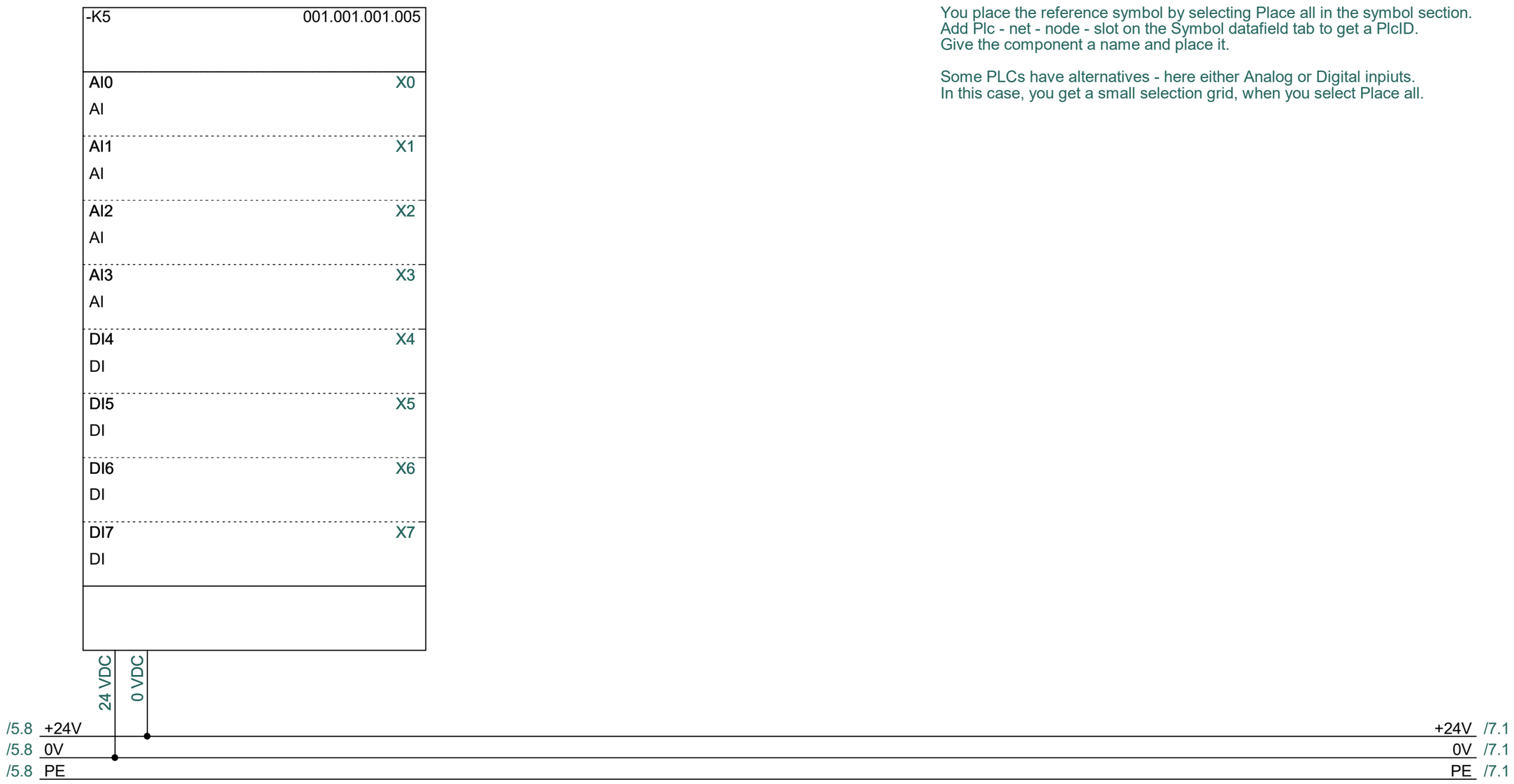
/4.8 +24V +24V /6.1
/4.8 0V 0V /6.1
/4.8 PE PE /6.1

You place the reference symbol by selecting Place all in the symbol section.
Add Plc - net - node - slot on the Symbol datafield tab to get a PlcID.
Give the component a name and place it.

All PLCs are pre-addressed from the database - Digital addresses are DI/DO and analog addresses are AI/AO.
Some components are defined with a statusype, which shows on the ref.symbol and on the IO-symbol.

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You place the reference symbol by selecting Place all in the symbol section.
Add Plc - net - node - slot on the Symbol datafield tab to get a PlcID.
Give the component a name and place it.

Some PLCs have alternatives - here either Analog or Digital inpiuts.
In this case, you get a small selection grid, when you select Place all.

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-K6	002.001.001.001		
DI0	X0	DO0	Y0
DI		DO	
DI1	X1	DO1	Y1
DI		DO	
DI2	X2	DO2	Y2
DI		DO	
DI3	X3	DO3	Y3
DI		DO	
DI4	X4	DO4	Y4
DI		DO	
DI5	X5	DO5	Y5
DI		DO	
DI6	X6	DO6	Y6
DI		DO	
DI7	X7	DO7	Y7
DI		DO	

+ ' PE

/6.8 +24V — +24V /8.1

/6.8 0V — 0V /8.1

/6.8 PE — PE /8.1

You place the reference symbol by selecting Place all in the symbol section.
Add Plc - net - node - slot on the Symbol datafield tab to get a PlcID.
Give the component a name and place it.

More than eight addresses means more columns, here two columns.

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-K7	002.001.001.002						
DI0 DI	X0 I0	DI10 DI Sink	X8 I8	DO0 DO	Y0 O0	DO10 DO Relay	Y8 O8
DI1 DI	X1 I1	DI11 DI Sink	X9 I9	DO1 DO	Y1 O1	DO11 DO Relay	Y9 O9
DI2 DI	X2 I2	DI12 DI Sink	X10 I10	DO2 DO	Y2 O2	DO12 DO Relay	Y10 O10
DI3 DI	X3 I3	DI13 DI Sink	X11 I11	DO3 DO	Y3 O3	DO13 DO Relay	Y11 O11
DI4 DI	X4 I4	DI14 DI Sink	X12 I12	DO4 DO	Y4 O4	DO14 DO Relay	Y12 O12
DI5 DI	X5 I5	DI15 DI Sink	X13 I13	DO5 DO	Y5 O5	DO15 DO Relay	Y13 O13
DI6 DI	X6 I6	DI16 DI Sink	X14 I14	DO6 DO	Y6 O6	DO16 DO Relay	Y14 O14
DI7 DI	X7 I7	DI17 DI Sink	X15 I15	DO7 DO	Y7 O7	DO17 DO Relay	Y15 O15

+ ' PE

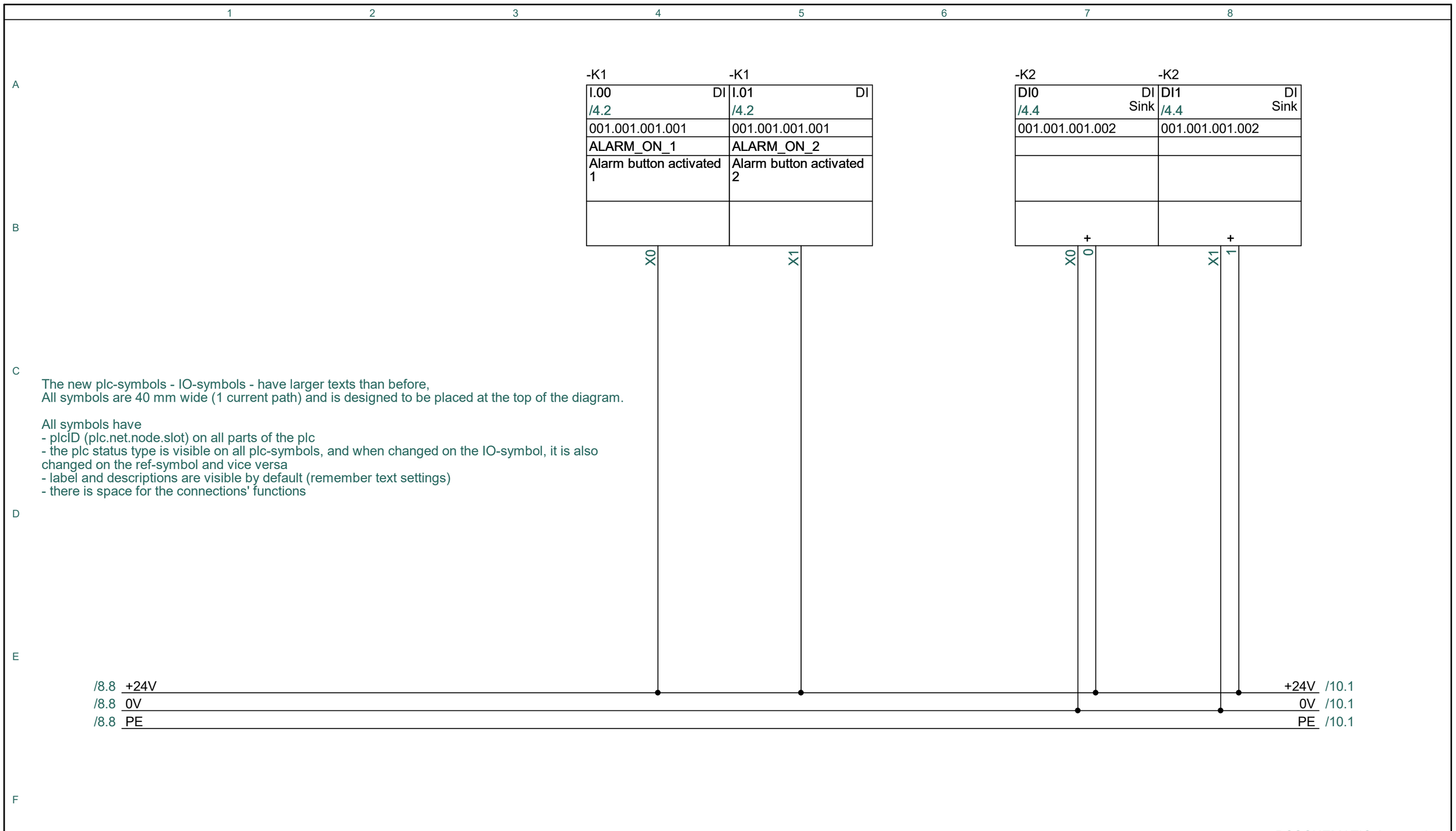
/7.8 +24V — +24V /9.1

/7.8 0V — 0V /9.1

/7.8 PE — PE /9.1

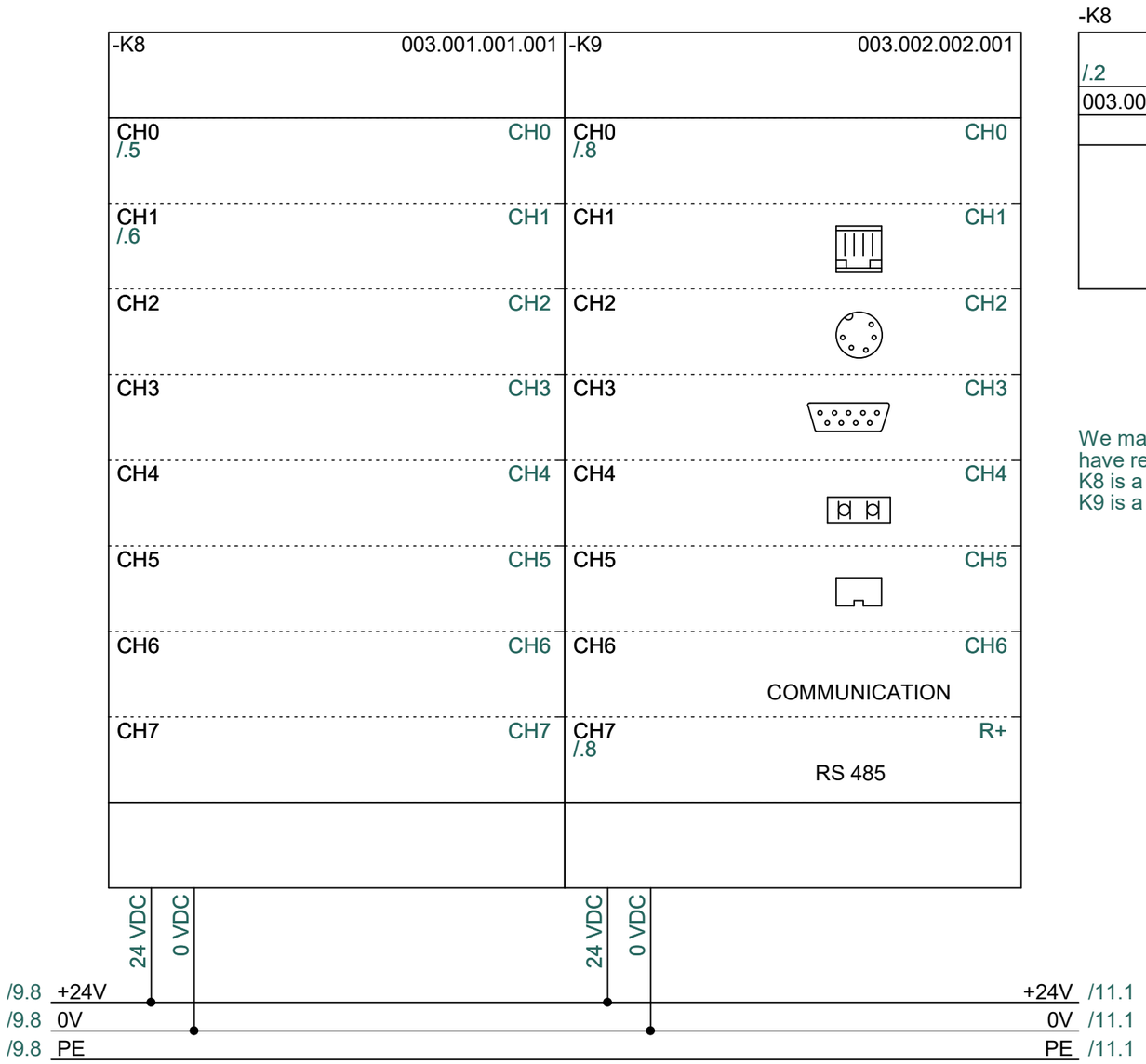
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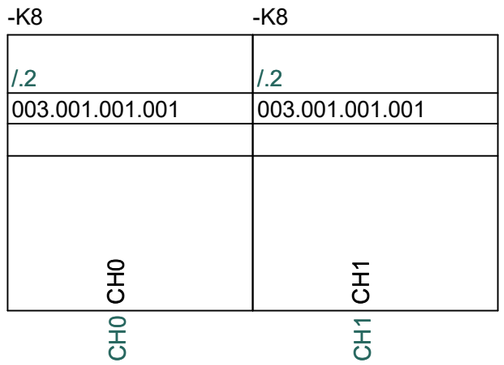
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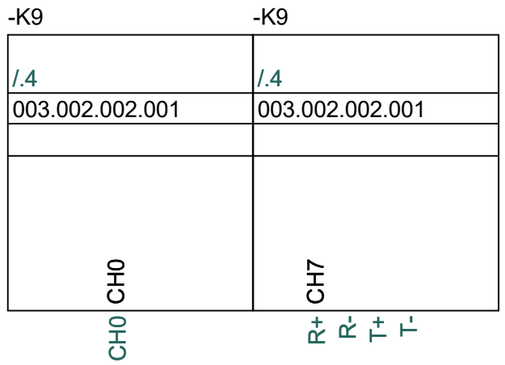


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We make comm.modules in the same way as we make plcs, meaning that they also have reference symbols and external symbols.
K8 is a component where all channels are the same
K9 is a component with eight different channel types

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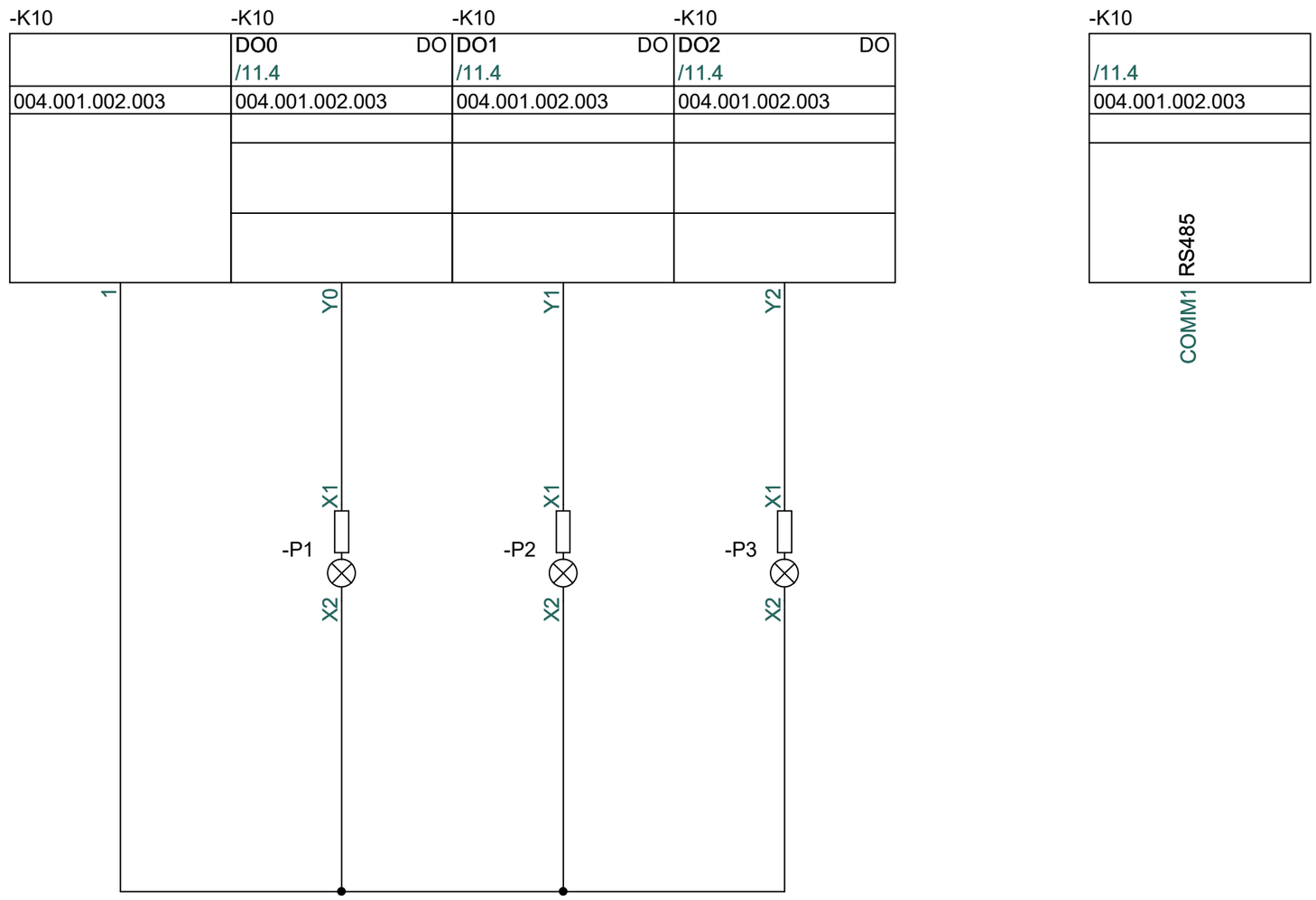
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-K10	004.001.002.003		
DI0 DI	X0	DO0 /12.2 DO	Y0
DI1 DI	X1	DO1 /12.3 DO	Y1
DI2 DI	X2	DO2 /12.4 DO	Y2
DI3 DI	X3	DO3 DO	Y3
AI1 AI 4-20mA	X4 4	DO4 DO	Y4
AI2 AI 4-20mA	X5 5	DO5 DO	Y5
AI3 AI 4-20mA	X6 6	RS485 /12.6	COMM1
AI4 AI 4-20mA	X7 7	COMM COMMUNICATION	COMM2

The reference card contains Inputs, Outputs and Communications.

/10.5	+24V	+24V
/10.5	0V	0V
/10.5	PE	PE

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Layout

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

A
B
C
D
E
F
G
H
I
J
K
L
M
N
O

Component	Position	Type	Function

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Lists

Component	Article no.	Type	Function	Description	Manufacturer	Position
-K1	PCS8920001	PCS-PLC-Card1		PLC-card, 8 Digital inputs, type1 (DI)	PCSCHEMATIC	/4.1
-K2	PCS8920002	PCS-PLC-Card2		PLC-card 8, Digital inputs, type2 (Sink)	--	/4.3
-K3	PCS8920003	PCS-PLC-Card3		PLC-card, 8 Digital output, type1 (DO)	--	/5.1
-K4	PCS8920004	PCS-PLC-Card4		PLC-card, 8 Digital output, type2 (Relay)	--	/5.3
-K5	PCS8920005	PCS-PLC-Card5		PLC-card, 8 inputs (DI/AI)	--	/6.1
-K6	PCS8920101	PCS-PLC-Com1-1		Compact plc 8 inputs, 8 outputs, type1	--	/7.1
-K7	PCS8920201	PCS-PLC-Com2-1		Compact plc 16 inputs, 16 outputs (digital)	--	/8.1
-K8	PCS8920501	PCS-Comm-1		Communications module - 8 channels	--	/10.1
-K9	PCS8920502	PCS-Comm2		Communications module, 8 different channel types	--	/10.3
-K10	PCS8920301	PCS-PLC-Com3-1		Compact plc 8 inputs, 6 outputs, 2 comm type1	--	/11.1
-P1	PCS500001	PCS-RDLAMP1		Red lamp with built-in resistor	--	/12.2
-P2	PCS500002	PCS-YELAMP1		Yellow lamp with built-in resistor	--	/12.3
-P3	PCS500003	PCS-GNLAMP1		Green lamp with built-in resistor	--	/12.4

Name	PlcID	Adress	Signal name	Description	Connected via	Conn. component	Position
-K1:X0	001.001.001.001	I.00	ALARM_ON_1	Alarm button activated 1		-K1	/9.4
-K1:X1	--	I.01	ALARM_ON_2	Alarm button activated 2		-K2	/9.5
-K1:X2	--	I.02	ALARM_ON_3	Alarm button activated 3			/4.2
-K1:X3	--	I.03	ALARM_ON_4	Alarm button activated 4			/4.2
-K1:X4	--	I.04	ALARM_ON_5	Alarm button activated 5			/4.2
-K1:X5	--	I.05	ALARM_ON_6	Alarm button activated 6			/4.2
-K1:X6	--	I.06	ALARM_ON_7	Alarm button activated 7			/4.2
-K1:X7	--	I.07	ALARM_ON_8	Alarm button activated 8			/4.2
-K2:X0	001.001.001.002	DI0				-K2	/9.7
-K2:X1	--	DI1				-K2	/9.8
-K2:X2	--	DI2					/4.4
-K2:X3	--	DI3					/4.4
-K2:X4	--	DI4					/4.4
-K2:X5	--	DI5					/4.4
-K2:X6	--	DI6					/4.4
-K2:X7	--	DI7					/4.4
-K3:X0		DO0					/5.2
-K3:X1		DO1					/5.2
-K3:X2		DO2					/5.2
-K3:X3		DO3					/5.2
-K3:X4		DO4					/5.2
-K3:X5		DO5					/5.2
-K3:X6		DO6					/5.2
-K3:X7		DO7					/5.2
-K4:Y0		O0					/5.4

Name	PlcID	Adress	Signal name	Description	Connected via	Conn. component	Position
-K4:Y1		O1					/5.4
-K4:Y2		O2					/5.4
-K4:Y3		O3					/5.4
-K4:Y4		O4					/5.4
-K4:Y5		O5					/5.4
-K4:Y6		O6					/5.4
-K4:Y7		O7					/5.4
-K5:X0		AI0					/6.2
-K5:X1		AI1					/6.2
-K5:X2		AI2					/6.2
-K5:X3		AI3					/6.2
-K5:X4		DI4					/6.2
-K5:X5		DI5					/6.2
-K5:X6		DI6					/6.2
-K5:X7		DI7					/6.2
-K6:X0		DI0					/7.2
-K6:X1		DI1					/7.2
-K6:X2		DI2					/7.2
-K6:X3		DI3					/7.2
-K6:X4		DI4					/7.2
-K6:X5		DI5					/7.2
-K6:X6		DI6					/7.2
-K6:X7		DI7					/7.2
-K6:Y0		DO0					/7.4
-K6:Y1		DO1					/7.4
-K6:Y2		DO2					/7.4

Name	PlcID	Adress	Signal name	Description	Connected via	Conn. component	Position
-K6:Y3		DO3					/7.4
-K6:Y4		DO4					/7.4
-K6:Y5		DO5					/7.4
-K6:Y6		DO6					/7.4
-K6:Y7		DO7					/7.4
-K7:X0		DI0					/8.2
-K7:X1		DI1					/8.2
-K7:X2		DI2					/8.2
-K7:X3		DI3					/8.2
-K7:X4		DI4					/8.2
-K7:X5		DI5					/8.2
-K7:X6		DI6					/8.2
-K7:X7		DI7					/8.2
-K7:X8		DI10					/8.4
-K7:X9		DI11					/8.4
-K7:X10		DI12					/8.4
-K7:X11		DI13					/8.4
-K7:X12		DI14					/8.4
-K7:X13		DI15					/8.4
-K7:X14		DI16					/8.4
-K7:X15		DI17					/8.4
-K7:Y0		DO0					/8.6
-K7:Y1		DO1					/8.6
-K7:Y2		DO2					/8.6
-K7:Y3		DO3					/8.6
-K7:Y4		DO4					/8.6
-K7:Y5		DO5					/8.6

Name	PlcID	Adress	Signal name	Description	Connected via	Conn. component	Position
-K7:Y6		DO6					/8.6
-K7:Y7		DO7					/8.6
-K7:Y8		DO10					/8.8
-K7:Y9		DO11					/8.8
-K7:Y10		DO12					/8.8
-K7:Y11		DO13					/8.8
-K7:Y12		DO14					/8.8
-K7:Y13		DO15					/8.8
-K7:Y14		DO16					/8.8
-K7:Y15		DO17					/8.8
-K10:X0	004.001.002.003	DI0					/11.2
-K10:X1	--	DI1					/11.2
-K10:X2	--	DI2					/11.2
-K10:X3	--	DI3					/11.2
-K10:X4	--	AI1					/11.2
-K10:X5	--	AI2					/11.2
-K10:X6	--	AI3					/11.2
-K10:X7	--	AI4					/11.2
-K10:Y0	--	DO0			-P1		/12.2
-K10:Y1	--	DO1			-P2		/12.3
-K10:Y2	--	DO2			-P3		/12.4
-K10:Y3	--	DO3					/11.4
-K10:Y4	--	DO4					/11.4
-K10:Y5	--	DO5					/11.4